

Chapter 8

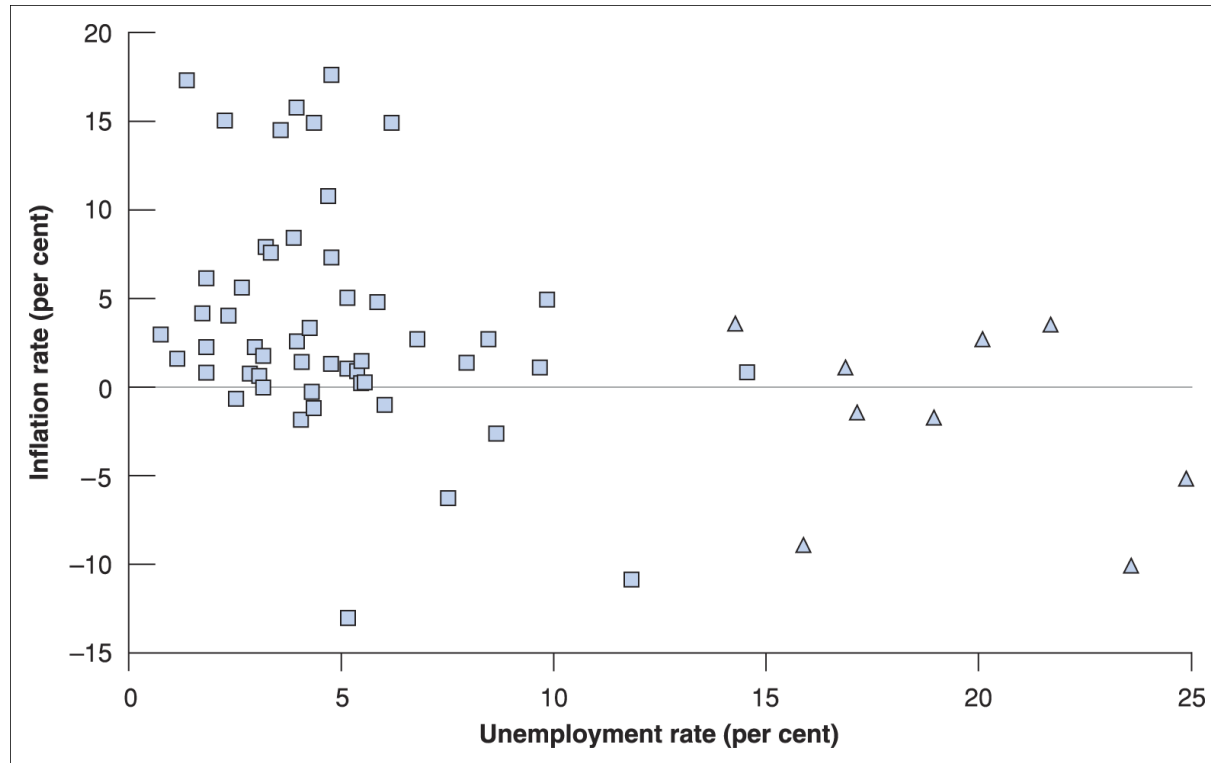
The Phillips curve

The Phillips curve

- In 1958, Phillips drew a diagram plotting the **rate of inflation** against the **rate of unemployment** in the UK for each year from 1861 to 1957
- He found clear evidence of a **negative relation** between **inflation** and **unemployment**
- In 1960, Samuelson and Solow replicated the exercise of the US from 1900 to 1960
- They also found evidence of a negative relation and labelled it the **Phillips curve**
- In the 1970s the relation broke down but a **new negative relation** between the **change in inflation** and **unemployment** emerged

The US data *before* 1970

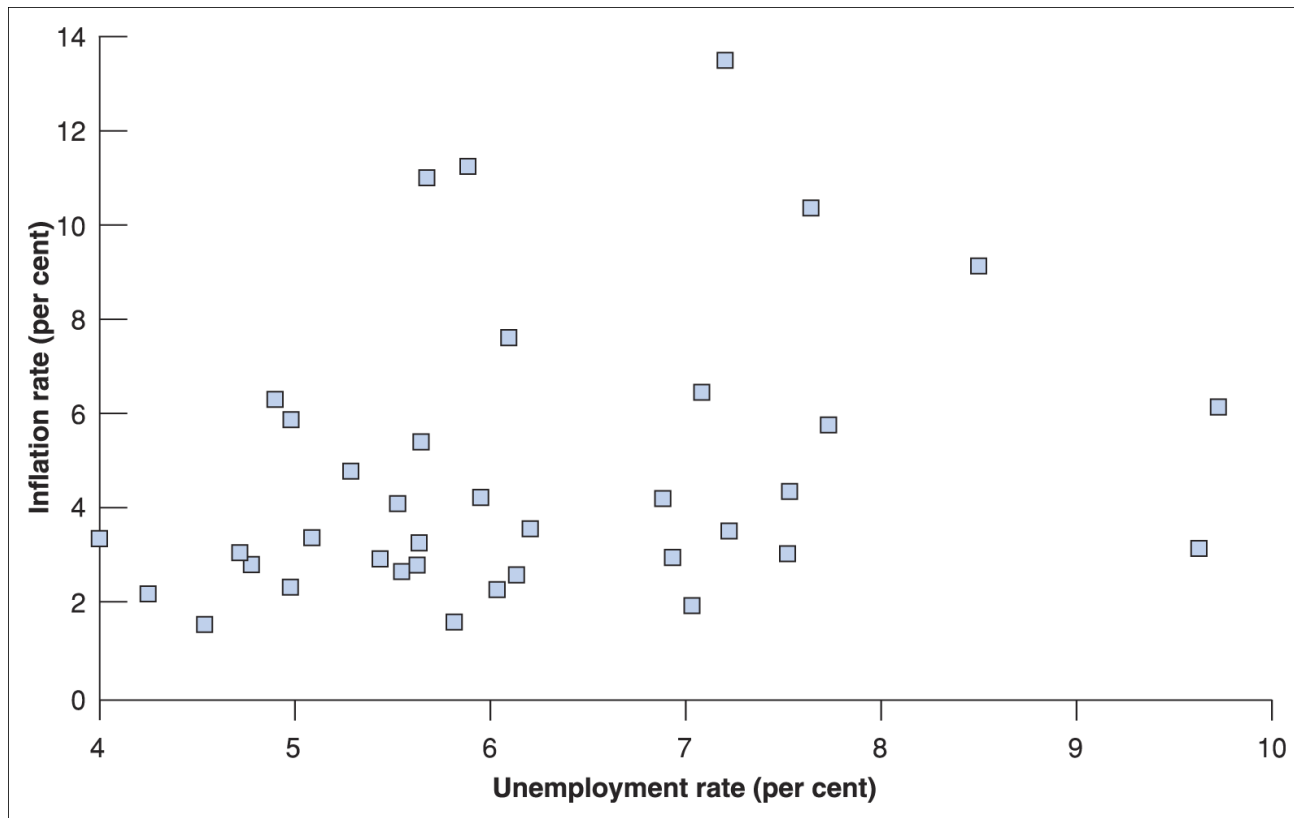
Inflation versus unemployment in the US, 1900–1960



During the period 1900-1960 in the US, a low unemployment rate was typically associated with a high inflation rate, and a high unemployment rate was typically associated with a low or negative inflation rate

The US data *after* 1970

Inflation versus unemployment in the US since 1970

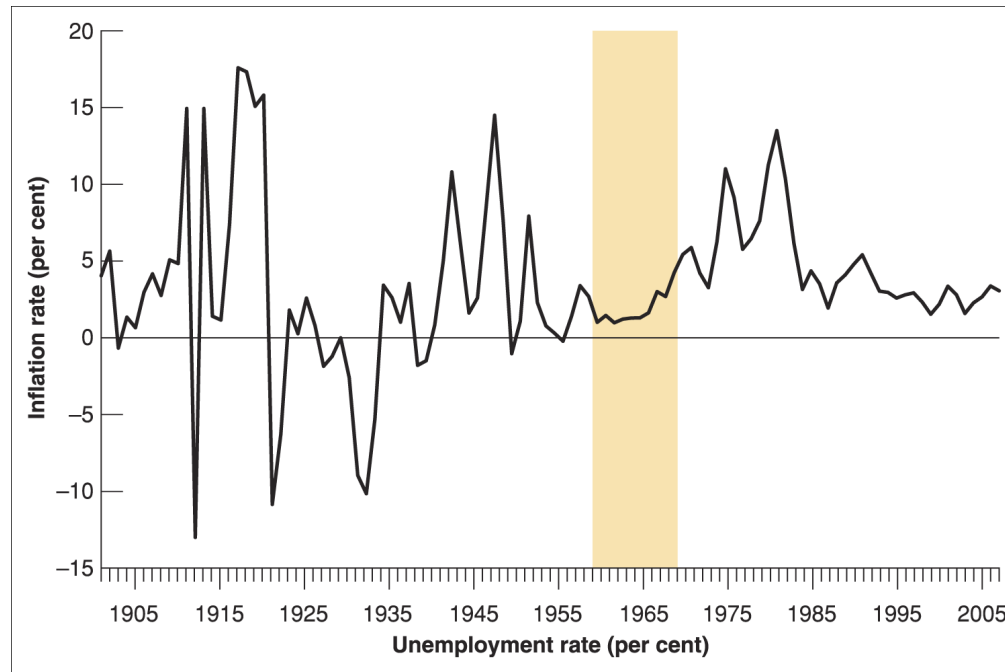


Beginning in 1970 in the United States, the relation between the unemployment rate and the inflation rate disappeared

After 1970, the relationship vanishes... why?

Because of a **change in the way wage-setters form expectations** due to a change in the behavior of inflation

US inflation from 1900 to 2005



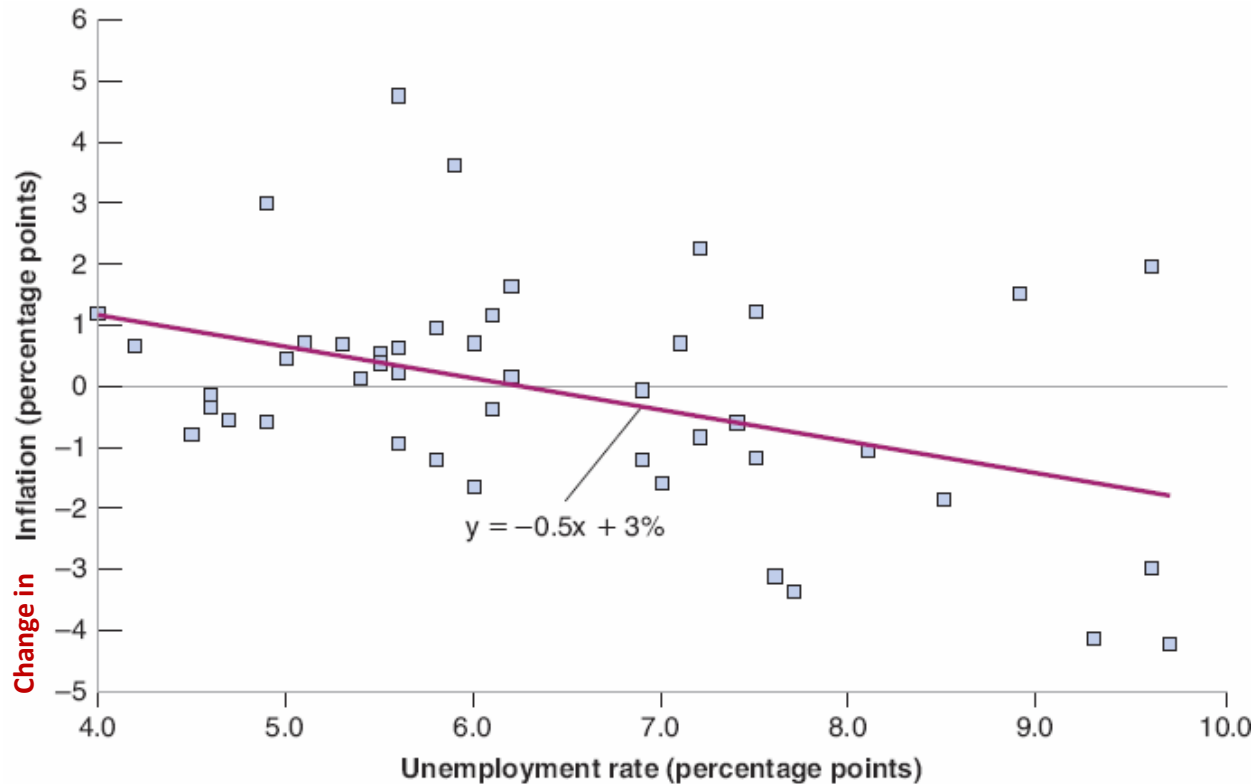
Before 1970 **inflation** is at times positive and at times negative, **not very persistent**

After 1970 **inflation** is always positive and **persistent**

- Why is this important? Because workers need to predict prices when they bargain for the wage
- If inflation is always positive and persistent, then workers will expect prices to rise in the future... (more on this later)

The US data *after* 1970

Change in inflation versus unemployment in the US since 1970



While the relation between inflation and unemployment broke down in the 1970s, a new relation between changes in inflation and unemployment reappeared

Plan

- Derive the relation between inflation and unemployment: the Phillips curve
- Explain the mutations of the Phillips curve over time
- Show the relation btw the Phillips curve and the natural unemployment rate
- More generally, explain the relation between inflation and unemployment

The Phillips curve equation: derivation

$$\text{PS} \quad P = (1 + m)W \quad W = P^e F(u, z) \quad \text{WS}$$



$$P = (1 + m)P^e F(u, z)$$

The **price, P** , depends on the **expected price, P^e** , and the **unemployment rate, u** (and also on m and z)

Implicit assumption: **prices** set by firms are **flexible**; **wages** set by workers and firms are **sticky**

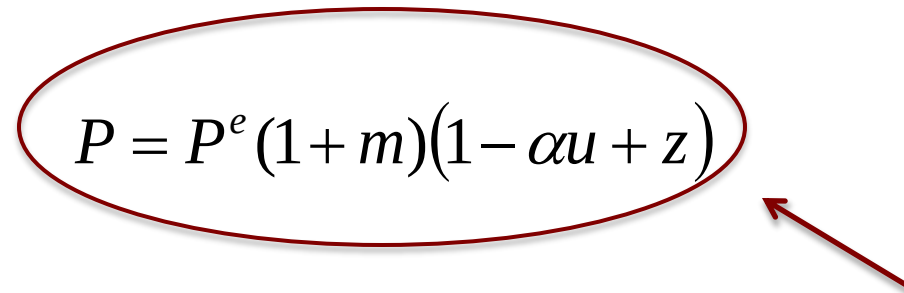
- We have

$$P = P^e(1+m)F(u,z)$$

- Now assume that (recall macroeconomists like linear relations!)

$$F(u, z) = 1 - \alpha u + z$$

- Combine


$$P = P^e(1+m)(1 - \alpha u + z)$$

**Aggregate supply relation from
price and wage setting**

- Add time indexes

$$P_t = P_t^e (1 + m)(1 - \alpha u_t + z)$$

- Divide both sides by the price level at t-1

$$\frac{P_t}{P_{t-1}} = \frac{P_t^e}{P_{t-1}} (1 + m)(1 - \alpha u_t + z)$$

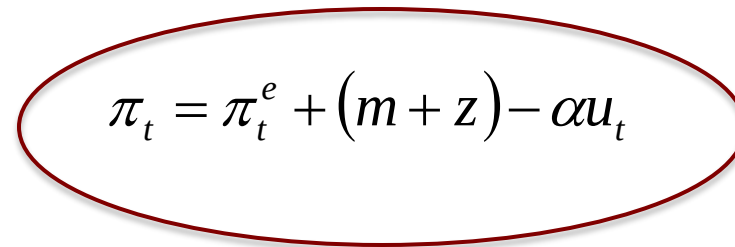
- Rewrite using the definition of inflation

$$(1 + \pi_t) = (1 + \pi_t^e)(1 + m)(1 - \alpha u_t + z)$$

- We have obtained

$$(1 + \pi_t) = (1 + \pi_t^e)(1 + m)(1 - \alpha u_t + z)$$

- This is approximately equal to


$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

Phillips curve



Price and wage setting: economic intuition

$$P_t = P_t^e (1 + m)(1 - \alpha u_t + z)$$

- A **higher expected price P_t^e** causes a **higher price P_t**

Higher expected price leads to higher wage (workers care about real wages)

Higher wage leads to higher price (markup over marginal cost)

- A **lower unemployment rate u_t** causes a **higher price P_t**

Lower unemployment leads to higher wage (bargaining)

Higher wage leads to higher price (markup over marginal cost)

$$P_t = P_t^e (1 + m)(1 - \alpha u_t + z)$$

- A **higher z** causes a **higher price P_t**

A higher z raises wages (bargaining)

Higher wages lead to higher prices (markup over marginal costs)

- A **higher markup m** causes a **higher price P_t**

Firms charge higher prices for given wages

The Phillips curve equation: economic intuition

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

- When workers expect higher inflation π_t^e , then inflation π_t is higher. Why?

Workers bargain higher nominal wages → firms set higher prices → inflation this period is higher *for given price last period*

- When the markup m is higher, then inflation π_t is higher. Why?

Higher markup → firms charge higher prices for given wages → inflation this period is higher *for given price last period*

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

- When z is higher then inflation π_t is higher. Why?

Workers bargain higher wages → firms set higher prices → inflation this period is higher *for given price last period*

- When the unemployment rate u_t is higher, then inflation π_t is lower. Why?

Workers bargain lower wages → firms set lower price → inflation this period is higher *for given price last period*

Note that the **parameter α** captures the strength of the effect of unemployment on bargained wages

The Phillips curve and its mutations

The early incarnation

- We saw that before 1970 Inflation was sometimes positive, sometimes negative, and **not very persistent**
- If inflation is not very persistent, then inflation last year is not a good predictor of inflation this year
- Assume inflation varies from year to year **around some average value** $\bar{\pi}$ and is not very persistent
- Then, it makes sense for wage setters to assume that, **whatever inflation was last year, inflation this year will simply be equal to its average value:**

$$\pi_t^e = \bar{\pi}$$

- The Phillips curve before the 1970s can then be written as

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t$$

- This is precisely the negative relation between unemployment and inflation holding until the 1970s

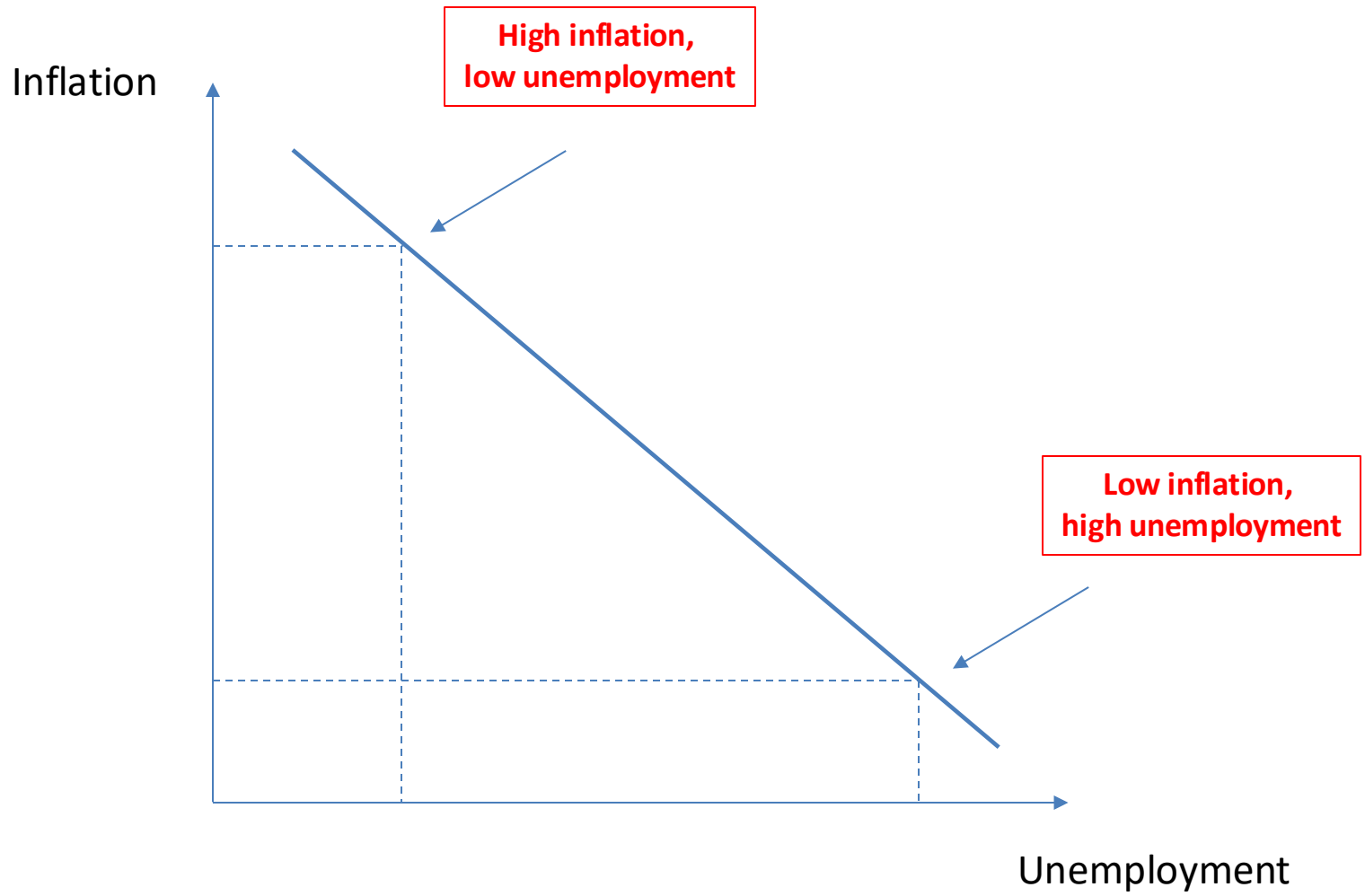
The Phillips curve and its mutations

The apparent trade-off btw inflation and unemployment

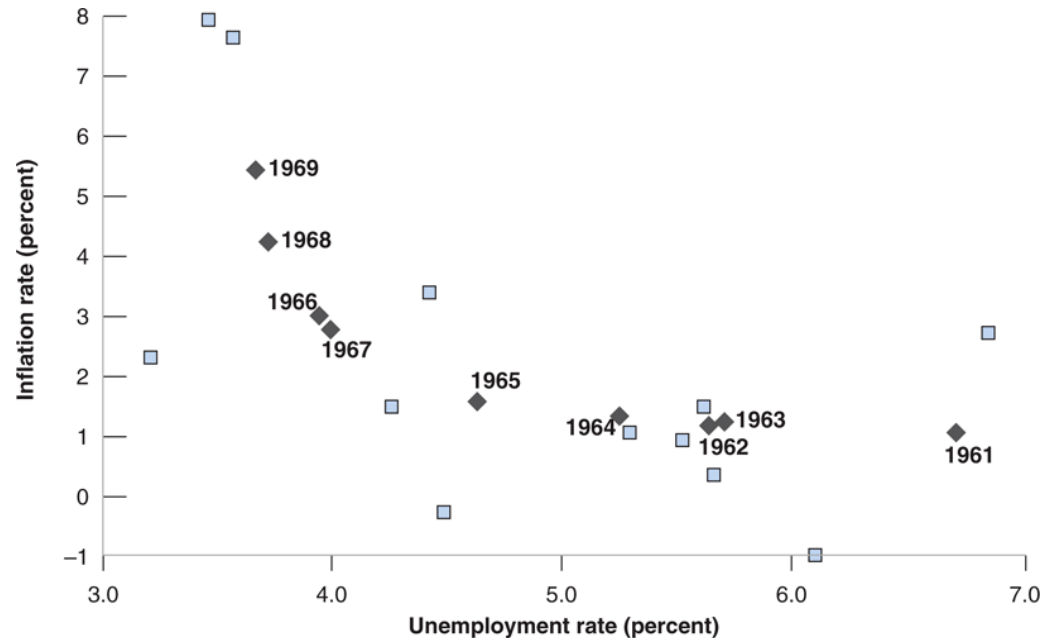
$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t$$

This relation has a striking implication:

A country can sustain a low unemployment rate if it is willing to tolerate a high inflation rate; vice versa, a country can achieve a low inflation rate as long as it is willing to tolerate a high unemployment rate; it just needs to choose a point on the Phillips curve



Inflation versus unemployment in the US from 1948 to 1969



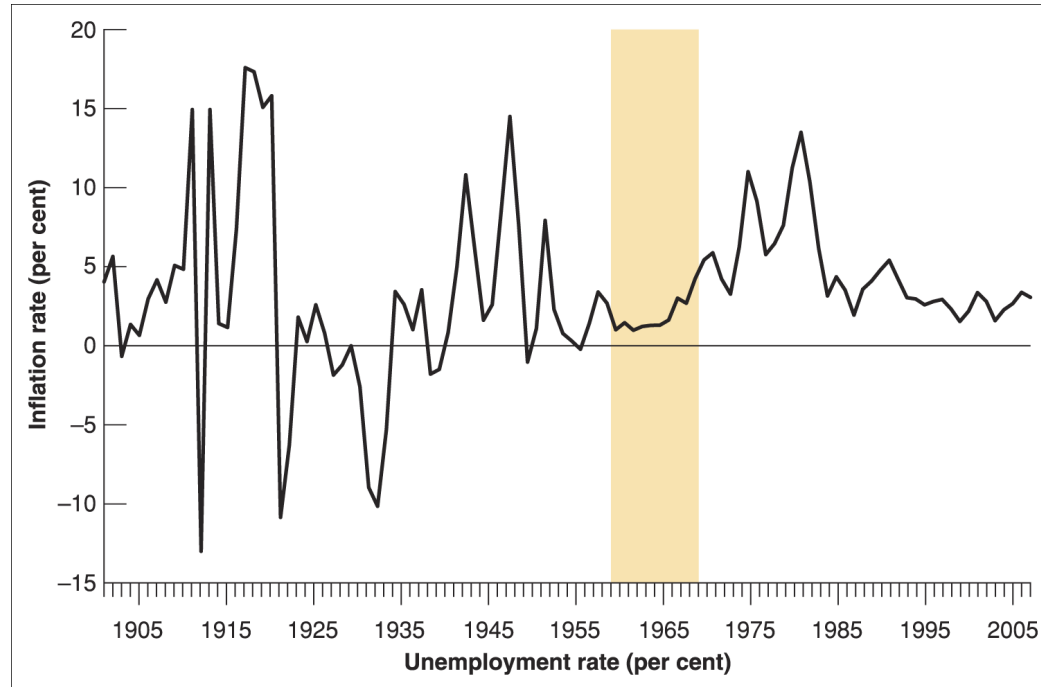
Source: Series UNRATE,
CPIAUSCL Federal Reserve Eco-
nomic Data (FRED) <http://research.stlouisfed.org/fred2/>

The steady decline in the U.S. unemployment rate throughout the 1960s was associated with a steady increase in the inflation rate

The Phillips curve and its mutations

The disappearance of the trade-off

What changed after 1970? A **change** in the **way wage setters formed inflation expectations**



After 1970 **inflation** is always positive and **more persistent**

- Workers need to predict prices when they bargain for the wage
- If inflation is always positive and persistent, then workers will expect prices to rise in the future and bargain for higher wages

Suppose workers look at the past year when forming expectations:

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

- Expected inflation this year depends partly on a constant average value, with weight $(1-\theta)$, and partly on inflation last year, with weight θ
- The higher is θ the more inflation last year leads workers to revise their expectations of what inflation will be this year

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

- We can think of what happened in the 1970s as **an increase in the value of θ over time**
- **Before the 1970s: $\theta = 0$** , workers expected inflation to be constant, whatever last year inflation
- **After the 1970s: $\theta = 1$** , workers started to expect inflation to be the same as last year

The Phillips curve then becomes

$$\pi_t = \underbrace{(1 - \theta)\bar{\pi} + \theta\pi_{t-1}}_{\text{Expected inflation}} + (m + z) - \alpha u_t$$

To distinguish it from the Phillips curve without expectations, this is often called the **expectations-augmented Phillips curve**

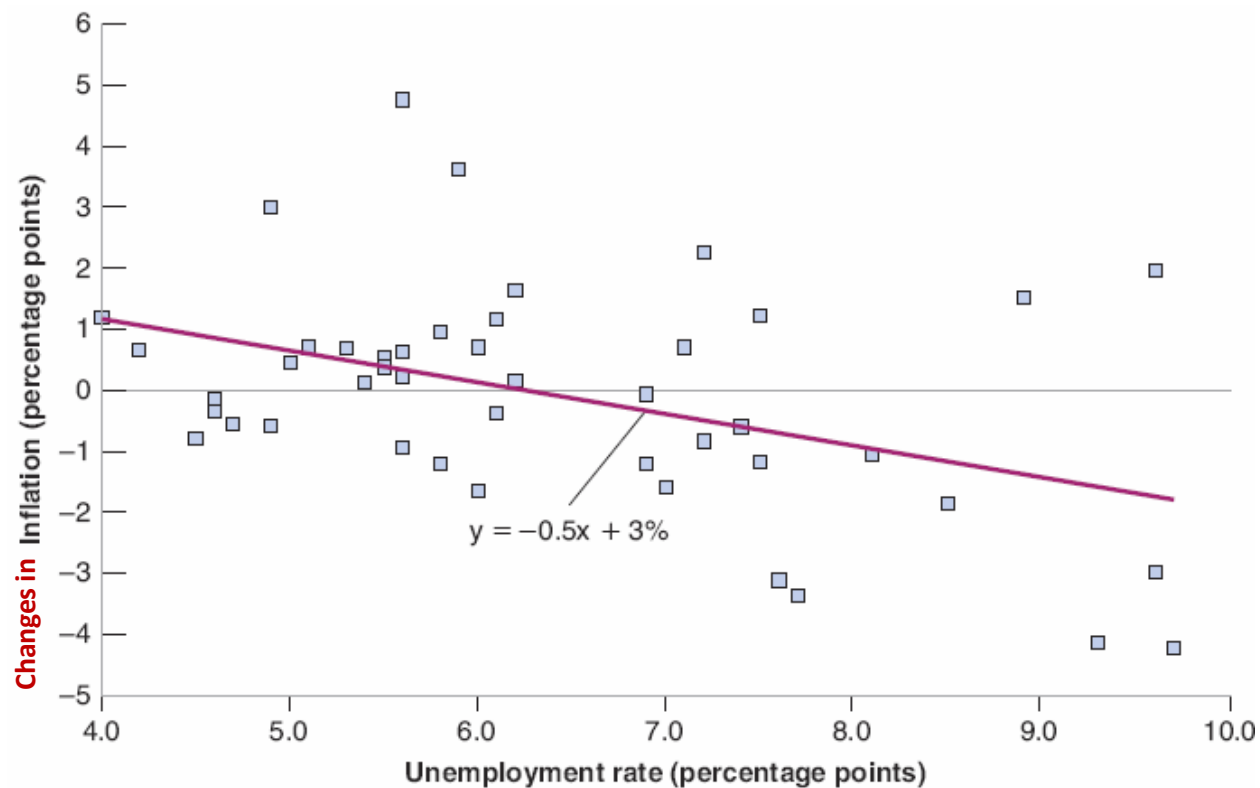
- When $\theta = 1$, the relation becomes

$$\pi_t = \underbrace{\pi_{t-1}}_{\text{Expected inflation rate}} + (m + z) - \alpha u_t$$

- Moving last year inflation rate to the left side of the equation, we get

$$\underbrace{\pi_t - \pi_{t-1}}_{\text{Change in the inflation rate}} = (m + z) - \alpha u_t$$

- Then, when $\theta = 1$, the unemployment rate affects **not** the **inflation rate**, **but** the **change in the inflation rate**
- This is the negative relation emerging between the **unemployment rate** and the **change in inflation** since the 1970s



The line that best fits the scatter of points for the period 1970–2014 is:

$$\pi_t - \pi_{t-1} = 3.0\% - 0.5u_t$$

The Phillips curve and the natural rate of unemployment

- The history of the Phillips curve is closely related to the concept of the **natural unemployment rate**
- The **original Phillips curve** implied that there is **no natural unemployment rate**: if policymakers were willing to tolerate a higher inflation rate, they could maintain a lower unemployment rate forever
- In the 1960s two economists, Friedman and Phelps, predicted that **the trade-off between inflation and unemployment would disappear** and unemployment could not be sustained below a certain level, which they called the natural unemployment rate. **They were right!**
- Let's now make explicit the relation between the Phillips curve and the natural unemployment rate

- Remember: the natural unemployment rate is the level at which **expectations** are **correct** (workers get the inflation right)
- So, the **natural unemployment rate** emerges when $\pi_t = \pi_t^e$
- Thus, the Phillips curve

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

becomes

$$0 = (m + z) - \alpha u_n$$

which implies

$$u_n = \frac{m + z}{\alpha}$$

$$u_n = \frac{m + z}{\alpha}$$

We can put this back into the Phillips curve to get

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$



$$\pi_t = \pi_t^e + \alpha u_n - \alpha u_t$$



A red arrow pointing from the left towards the equation.
$$\pi_t - \pi_t^e = -\alpha(u_t - u_n)$$

**The Phillips curve in terms of
the natural rate of unemployment**

If $\theta = 1$ and expected inflation equals last year inflation we have

$$\pi_t - \underbrace{\pi_{t-1}}_{\text{Expected inflation}} = -\alpha(u_t - u_n)$$

- when u is lower than u_n , the inflation rate increases
- when u is higher than u_n , the inflation rate decreases
- when u equals u_n the inflation rate is constant (this is why the natural rate of unemployment is also called the non-accelerating inflation rate of unemployment - NAIRU)

Back to the natural rate of unemployment:

$$u_n = \frac{m + z}{\alpha}$$

Factors raising u_n :

- **Labor market rigidities (high z):**
 - high minimum wages
 - high unemployment benefits
 - strict employment protection legislation (high layoffs costs)
- **Low degree of product market competition (high m)**

- The factors that affect the natural rate of unemployment, m and z , **differ across countries** and **over time**
- There is no reason to expect all countries to have the same natural rate of unemployment
- The natural rate of unemployment of one country may also vary over time as m and z may vary, though slowly

- The natural rate of unemployment is higher in Europe than in the US
- In Europe the natural rate of unemployment has greatly increased since the 1960s
- In the US the natural rate of unemployment increased from the 1960s to the 1980s and appears to have decreased since then

The natural rate of unemployment in some European countries

Country	1970–2008	1970–1990	1991–2008
Denmark	6.0%	5.8%	6.5%
Finland	77.0%	4.6%	9.3%
France	7.4%	6.5%	9.4%
Germany	6.0%	4.1%	8.6%
Ireland	7.5%	9.0%	9.0%
Netherlands	5.6%	6.9%	4.7%
Sweden	3.9%	2.6%	7.3%
UK	6.7%	6.7%	6.7%

Source: OECD *Economic Outlook* database.

High inflation, wage indexation and the Phillips curve

- So far, workers form expectations about inflation and then wait for their next chance to renegotiate their contracts
 - When inflation is higher than expected, wages are too low and they ask for more
 - When inflation is lower than expected, wages are too high and employers renegotiate a lower wage
- But what if inflation is **high and volatile**?
 - Then workers and firms become reluctant to enter into labor contracts that set nominal wages for long time, as workers (firms) may experience large losses if inflation turns out to be much higher (lower) than expected
 - **Wage indexation** becomes prevalent

Wage indexation

- Wage indexation allows wages to adjust to inflation automatically (no need to bargain)
- Let's fit this into the Phillips curve:
 - Assume that a **fraction λ** of contracts is **automatically renegotiated**
 - So every year a fraction λ of wages **moves together with inflation**

The Phillips curve with wage indexation

- The Phillips curve with indexation becomes:

$$\pi_t = \underbrace{\lambda \overbrace{\pi_t}^{\text{current inflation}}}_{\text{indexed wages}} + \underbrace{(1-\lambda) \overbrace{\pi_t^e}^{\text{expected inflation}}}_{\text{regular wages}} - \alpha(u_t - u_n)$$

- Assume

$$\pi_t^e = \pi_{t-1}$$

- Then obtain

$$\pi_t = \lambda \pi_t + (1-\lambda) \pi_{t-1} - \alpha(u_t - u_n)$$

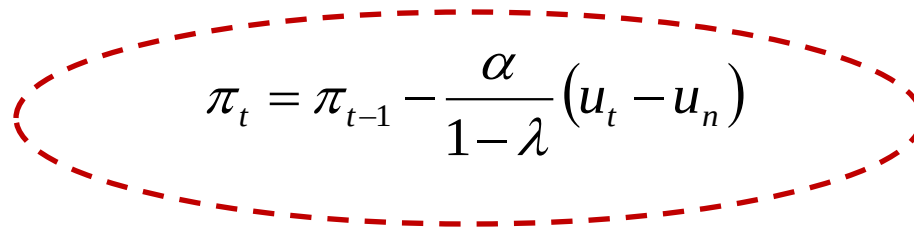
- We have

$$\pi_t = \lambda \pi_t + (1 - \lambda) \pi_{t-1} - \alpha (u_t - u_n)$$

- This can be rearranged as

$$(1 - \lambda) \pi_t = (1 - \lambda) \pi_{t-1} - \alpha (u_t - u_n)$$

- And finally as


$$\pi_t = \pi_{t-1} - \frac{\alpha}{1 - \lambda} (u_t - u_n)$$

Phillips curve with indexed wages

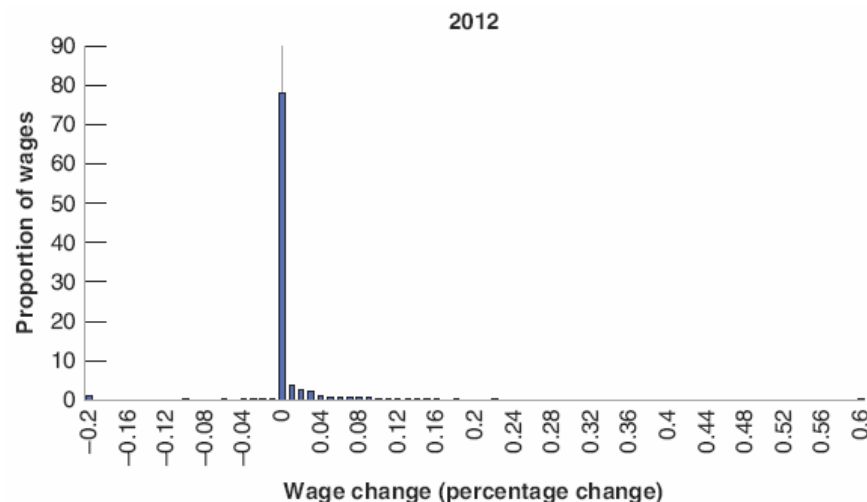
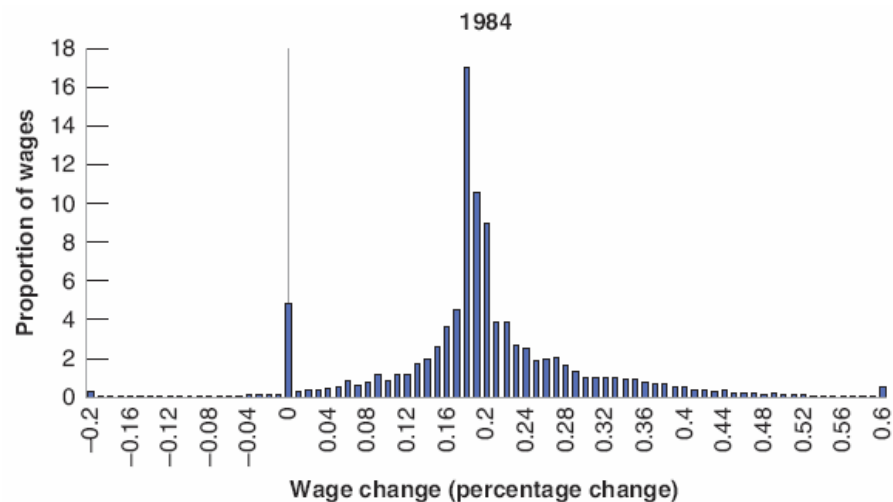
$$\pi_t = \pi_{t-1} - \frac{\alpha}{1-\lambda} (u_t - u_n)$$

- When a high fraction of wages is indexed (λ is high)...
- ... a change in unemployment has a **bigger effect on inflation**
- Why?
 - Without wage indexation, when unemployment decreases, wages raise and prices as well; but then, because wages do not respond to prices right away (need to revise expectations and renegotiate), there is no further increase in prices within the year
 - With wage indexation, an increase in prices leads to a further increase in wages (some workers get a raise automatically, with no bargaining), which in turn leads to a further increases in wages and so on...

Deflation and the Phillips curve

- When low inflation or deflation occurs, the Phillips curve breaks down
- One possible reason is the reluctance of workers to accept cuts in nominal wages: **downward nominal wage rigidity**
- This has happened in the US during the Great Depression and, to a lesser extent, during the Great Recession in a number of countries:
 - Given the high unemployment rate, the inflation rate was surprisingly high
 - Put it differently, given the high unemployment rate, we would have expected a large deflation rate, while deflation was limited

Distribution of real wage changes in Portugal, in times of high and low inflation



- In 1984, inflation was 27%, and the distribution of real wage changes was roughly symmetric
- In 2012, the inflation rate was just 2.1%, and the distribution of real wage changes was bunched at zero, with nearly no negative real wage changes